# Jenkins Installation on Ubuntu (EC2)

GPG Key Verification Error — Troubleshooting Guide

## Overview

This guide documents a common GPG signature verification error encountered when installing Jenkins on Ubuntu EC2 instances. It explains the root cause and provides step-by-step resolution commands.

## The Error

When running sudo apt update after adding the Jenkins repository, you may see:

```
W: GPG error: https://pkg.jenkins.io/debian-stable binary/ Release:
   The following signatures couldn't be verified because the public key
   is not available: NO_PUBKEY 7198F4B714ABFC68

E: The repository 'https://pkg.jenkins.io/debian-stable binary/ Release'
   is not signed.
N: Updating from such a repository can't be done securely, and is
   therefore disabled by default.
```

## Root Cause

**Why does this happen?**

The wget -O command saves the Jenkins key as a raw ASCII-armored (.asc) file. However, apt requires keys in binary GPG format (.gpg). Using wget saves the armored text as-is, which apt cannot verify — causing the NO_PUBKEY error even though the file exists on disk.

The key ID in the error message — `7198F4B714ABFC68` — is the fingerprint of the Jenkins signing key. apt looks for this key in the configured keyring but cannot find a valid binary-format key to match against.

## Resolution

### Method 1: Re-download and Dearmor the Key (Recommended)

This method fixes the format issue by using gpg --dearmor to convert the key to binary format.

| | |
|---|---|
| 1 | **Remove the old incorrect key file**<br>sudo rm /usr/share/keyrings/jenkins-keyring.asc |
| 2 | **Download the key and convert it to binary format**<br>curl -fsSL https://pkg.jenkins.io/debian-stable/jenkins.io-2023.key \| sudo gpg --dearmor -o /usr/share/keyrings/jenkins-keyring.gpg |
| 3 | **Update the apt sources list to reference the new .gpg key**<br>echo "deb [signed-by=/usr/share/keyrings/jenkins-keyring.gpg] https://pkg.jenkins.io/debian-stable binary/" \| sudo tee /etc/apt/sources.list.d/jenkins.list > /dev/null |
| 4 | **Update package lists**<br>sudo apt update |
| 5 | **Install Jenkins**<br>sudo apt install jenkins -y |

## Method 2: Fetch Key Directly from Keyserver

Use this if Method 1 fails or if the downloaded key does not match the expected key ID.

```
# Step 1: Remove old files
sudo rm /usr/share/keyrings/jenkins-keyring.gpg
sudo rm /etc/apt/sources.list.d/jenkins.list

# Step 2: Fetch the exact key from Ubuntu keyserver
sudo gpg --keyserver keyserver.ubuntu.com --recv-keys 7198F4B714ABFC68

# Step 3: Export key to keyrings directory
sudo gpg --export 7198F4B714ABFC68 | sudo tee /usr/share/keyrings/jenkins-
keyring.gpg > /dev/null

# Step 4: Re-add the repository
echo "deb [signed-by=/usr/share/keyrings/jenkins-keyring.gpg]
https://pkg.jenkins.io/debian-stable binary/" | sudo tee
/etc/apt/sources.list.d/jenkins.list > /dev/null

# Step 5: Update and install
sudo apt update
sudo apt install jenkins -y
```

## Method 3: Alternative Keyserver (if Ubuntu keyserver is unreachable)

```
# Try OpenPGP keyserver
sudo gpg --keyserver hkps://keys.openpgp.org --recv-keys 7198F4B714ABFC68
```

```
# OR fetch directly via curl
curl -fsSL
"https://keyserver.ubuntu.com/pks/lookup?op=get&search=0x7198F4B714ABFC68" \
  | sudo gpg --dearmor -o /usr/share/keyrings/jenkins-keyring.gpg
```

# Post-Installation Steps

## Start and Enable Jenkins Service

```
sudo systemctl start jenkins
sudo systemctl enable jenkins

# Check status
sudo systemctl status jenkins
```

## Get the Initial Admin Password

```
sudo cat /var/lib/jenkins/secrets/initialAdminPassword
```

**Access Jenkins UI**

Open your browser and navigate to: http://<your-ec2-public-ip>:8080 Make sure port 8080 is open in your EC2 Security Group inbound rules.

# Quick Reference

| Item | Detail |
|---|---|
| **Error Key ID** | `NO_PUBKEY 7198F4B714ABFC68` |
| **Wrong file format** | `.asc (ASCII-armored) — apt needs .gpg (binary)` |
| **Fix command** | `curl ... \| sudo gpg --dearmor -o ...keyring.gpg` |
| **Keyring location** | `/usr/share/keyrings/jenkins-keyring.gpg` |
| **Sources list** | `/etc/apt/sources.list.d/jenkins.list` |
| **Jenkins default port** | `8080` |
| **Initial password path** | `/var/lib/jenkins/secrets/initialAdminPassword` |